Name _____ Date _____ Period _____

Solving Linear Equations

Grade 8 · Linear Equations and Linear Systems

1. Solve $3(x - 4) = 12x$

2. Solve each of these equations. Explain or show your reasoning.

$2(x + 5) = 3x + 1$ $3y - 4 = 6 - 2y$ $3(n + 2) = 9(6 - n)$

3. Solve each equation, and check your solution.

$\frac{1}{9}(2m - 16) = \frac{1}{3}(2m + 4)$ $-4(r + 2) = 4(2 - 2r)$ $12(5 + 2y) = 4y - (6 - 9y)$

4. Solve each of these equations. Explain or show your reasoning.

$2b + 8 - 5b + 3 = -13 + 8b - 5$ $2x + 7 - 5x + 8 = 3(5 + 6x) - 12x$

$2c - 3 = 2(6 - c) + 7c$

5. Solve each equation and check your solution.

$-3w - 4 = w + 3$ $3(3 - 3x) = 2(x + 3) - 30$

$\frac{1}{3}(z + 4) - 6 = \frac{2}{3}(5 - z)$

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6. Andre solved an equation, but when he checked his answer he saw his solution was incorrect. He knows he made a mistake, but he can't find it. Where is Andre's mistake and what is the solution to the equation?

$$\begin{aligned} -2(3x-5) &= 4(x+3)+8 \\ -6x+10 &= 4x+12+8 \\ -6x+10 &= 4x+20 \\ 10 &= -2x+20 \\ -10 &= -2x \\ 5 &= x \end{aligned}$$

7. Clare was solving an equation, but when she checked her answer she saw her solution was incorrect. She knows she made a mistake, but she can't find it. Where is Clare's mistake and what is the solution to the equation?

$$\begin{aligned} 12(5+2y) &= 4y-(5-9y) \\ 72+24y &= 4y-5-9y \\ 72+24y &= -5y-5 \\ 24y &= -5y-77 \\ 29y &= -77 \\ y &= \frac{-77}{29} \end{aligned}$$

8. Elena said the equation $9x+15=3x+15$ has no solutions because $9x$ is greater than $3x$. Do you agree with Elena? Explain your reasoning.